

Database to UI Field Mapping Guide

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Overview

This document maps database fields from `shared/schema.ts` to their corresponding UI components in the ClientIQ banking application. It identifies which fields are displayed directly and which undergo calculations or transformations before display.

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Customer Data Mappings

Database Table: `customer` (`shared/schema.ts`, lines 100-163)
API Endpoint: `/api/customers/:customerId/details`
Primary Component: `CustomerOverview.tsx` , `CustomerDetailModal.tsx`

Direct Field Mappings

Database Column	UI Component	Display Element	Transformation
firstName	CustomerOverview	Name heading	Combined with lastName
lastName	CustomerOverview	Name heading	Combined with firstName
fullName	CustomerSearch	Search result item	Direct display
preferredName	CustomerOverview	Name (if exists)	Direct display
customerType	CustomerOverview	Type badge	Label mapping
customerStatus	CustomerOverview	Status chip	Color mapping (active=green)
taxIdentifier	CustomerOverview	Tax ID field	Masked display *****-**-xxxx (line 237)
jackHenryCifNumber	CustomerOverview	CIF Number	Direct display
silverlakeCustomerId	CustomerOverview	Silverlake ID	Direct display
customerSince	CustomerOverview	"Customer Since"	Date formatted
dateOfBirth	CustomerDetailModal	Date of Birth	Date formatted + Age calculated
vipCustomer	CustomerOverview	VIP badge	Boolean to badge
isEmployee	CustomerOverview	Employee badge	Boolean to badge
riskRating	RiskCompliance	Risk indicator	Color mapping (low/medium/high)
kycStatus	RiskCompliance	KYC status	Label mapping
kycLastUpdated	RiskCompliance	Last KYC date	Date formatted
occupation	CustomerDetailModal	Occupation field	Direct display
employerName	CustomerDetailModal	Employer field	Direct display
businessName	CustomerOverview	Name (for business)	Direct display

Database Column	UI Component	Display Element	Transformation
dbName	CustomerDetailModal	DBA field	Direct display
naicsCode	CustomerDetailModal	NAICS code	Direct display
branchId → branchName	CustomerOverview	Branch indicator	Joined with branch table, displayed as "Branch: {name}"
governmentId	CustomerDetailModal	Gov ID	Masked display
governmentIdType	CustomerDetailModal	ID type label	Direct display
maritalStatus	CustomerDetailModal	Marital status	Direct display
gender	CustomerDetailModal	Gender	Direct display
citizenship	CustomerDetailModal	Citizenship	Direct display
languagePreference	CustomerDetailModal	Language	Direct display

Calculated Fields

Derived Field	Source Fields	Component	Calculation
age	dateOfBirth	CustomerDetailModal, CustomerDashboard	calculateAge() function
displayName	firstName , lastName , businessName	CustomerOverview	Conditional concatenation
isBirthday	dateOfBirth	CustomerOverview	Compare month/day to today

Account Data Mappings

Database Table: account (shared/schema.ts, lines 188-219)
API Endpoint: /api/customers/:customerId/accounts
Primary Components: AccountSummaryTableVersion.tsx , AccountCard.tsx , AccountList.tsx

Direct Field Mappings

Database Column	UI Component	Display Element	Transformation
accountNumber	AccountSummaryTableVersion	Account number	Masked (****1234)
accountType	AccountSummaryTableVersion	Account type badge	Label mapping
accountSubtype	AccountCard	Subtype label	Direct display
accountStatus	AccountSummaryTableVersion	Status chip	Color mapping
balance	AccountSummaryTableVersion	Balance column	Currency formatted
availableBalance	AccountCard	Available balance	Currency formatted
currency	AccountCard	Currency indicator	Direct display
interestRate	AccountSummaryTableVersion	Rate column	Percentage formatted
creditLimit	AccountCard	Credit limit	Currency formatted
openedDate	AccountCard	Opened date	Date formatted
closedDate	AccountCard	Closed date	Date formatted
maturityDate	AccountCard	Maturity date	Date formatted (for CDs)
lastTransactionDate	AccountCard	Last activity	Date formatted
productCode	AccountCard	Product code	Direct display
jackHenryAccountId	AccountCard	JH Account ID	Direct display
silverlakeAccountStructure	AccountCard	SL Structure	Direct display
accountClass	AccountCard	Account class	Direct display
statementCycle	AccountCard	Statement cycle	Direct display
averageBalance	Deposits	Average balance	Currency formatted

Calculated Fields

Derived Field	Source Fields	Component	Calculation
productName	accountType , accountSubtype	AccountCard	getProductName() function
maskedAccountNumber	accountNumber	AccountCard	maskAccountNumber() - shows last 4
accountTypeLabel	accountType	AccountSummaryTableVersion	Lookup table mapping
accountIcon	accountType	AccountCard	getAccountIcon() - returns MUI icon

Transaction Data Mappings

Database Table: financialTransaction (shared/schema.ts, lines 378-419)
API Endpoints: /api/accounts/:accountId/transactions , /api/customers/:customerId/transactions
Primary Component: TransactionHistory.tsx

Direct Field Mappings

Database Column	UI Component	Display Element	Transformation
amount	TransactionHistory	Amount column	Currency formatted, +/- indicator
transactionCode	TransactionHistory	Code column	Direct display
transactionType	TransactionHistory	Type badge	Label mapping
status	TransactionHistory	Status chip	Color mapping
transactionDate	TransactionHistory	Date column	Date formatted
postingDate	TransactionHistory	Posting date	Date formatted
description	TransactionHistory	Description column	Direct display
referenceNumber	TransactionHistory	Reference	Direct display
merchantName	TransactionHistory	Merchant	Direct display
merchantCategoryCode	TransactionHistory	MCC	Direct display
ledgerBalanceAfter	TransactionHistory	Running balance	Currency formatted
availableBalanceAfter	TransactionHistory	Available	Currency formatted

Calculated Fields

Derived Field	Source Fields	Component	Calculation
totalCredits	All amount > 0	TransactionHistory	Sum aggregation
totalDebits	All amount < 0	TransactionHistory	Sum aggregation (abs)
transactionIcon	transactionType , categoryId	TransactionHistory	Icon mapping
formattedAmount	amount	TransactionHistory	formatCurrency() with +/- sign

Aggregation Code (TransactionHistory.tsx lines 164-172):

```
const totals = filteredTransactions.reduce((acc, trans) => {
  const amount = parseFloat(trans.amount);
  if (amount > 0) {
    acc.credits += amount;
  } else {
    acc.debits += Math.abs(amount);
  }
  return acc;
}, { credits: 0, debits: 0 });
```

Household Data Mappings

- Database Table:** household (shared/schema.ts, lines 165-186)
- Database Table:** householdMembership (shared/schema.ts, lines 248-268)
- API Endpoint:** /api/households/:id , /api/households/customer/:customerId
- Primary Components:** HouseholdRelationships.tsx , HouseholdPage.tsx

Direct Field Mappings

Database Column	UI Component	Display Element	Transformation
householdName	HouseholdPage	Page title	Direct display
householdType	HouseholdRelationships	Type badge	Label mapping
totalAssets	HouseholdPage	Total assets	Currency formatted
totalLiabilities	HouseholdPage	Total liabilities	Currency formatted
householdStatus	HouseholdPage	Status chip	Color mapping
riskRating	HouseholdPage	Risk indicator	Color mapping
establishedDate	HouseholdPage	Established date	Date formatted
taxFilingStatus	HouseholdPage	Tax status	Direct display
parentHouseholdId	HouseholdPage	Parent company link	Navigation link
consolidationMethod	HouseholdPage	Consolidation	Direct display

Membership Fields

Database Column	UI Component	Display Element	Transformation
relationshipRole	HouseholdRelationships	Role label	Direct display
isPrimaryMember	HouseholdRelationships	Primary badge	Boolean to badge
isHeadOfHousehold	HouseholdRelationships	Head badge	Boolean to badge
membershipStartDate	HouseholdRelationships	Member since	Date formatted
ownershipPercentage	HouseholdRelationships	Ownership %	Percentage formatted
controlType	HouseholdRelationships	Control type	Label mapping

Calculated Fields

Derived Field	Source Fields	Component	Calculation
netWorth	totalAssets , totalLiabilities	HouseholdPage	assets - liabilities
memberCount	membership records	HouseholdRelationships	Count of active members
totalHouseholdBalance	All member accounts	HouseholdPage	Sum of all account balances

Contact & Address Mappings

Database Tables: contactInfo , address , entityContact , entityAddress

API Endpoint: /api/customers/:customerId/contacts

Primary Component: ContactInformation.tsx

Contact Info Fields

Database Column	UI Component	Display Element	Transformation
contactType	ContactInformation	Type icon	Icon mapping
contactValue	ContactInformation	Contact display	Format by type (phone mask, email link)
contactSubtype	ContactInformation	Subtype label	Direct display
isPrimary	ContactInformation	Primary badge	Boolean to star icon
isVerified	ContactInformation	Verified icon	Boolean to checkmark
canContact	ContactInformation	Contact preference	Boolean indicator

Address Fields

Database Column	UI Component	Display Element	Transformation
addressLine1	ContactInformation	Address line 1	Direct display
addressLine2	ContactInformation	Address line 2	Direct display
city	ContactInformation	City	Direct display
state	ContactInformation	State	Direct display
postalCode	ContactInformation	ZIP	Direct display
country	ContactInformation	Country	Direct display
addressType	ContactInformation	Type label	Label mapping
isPrimary	ContactInformation	Primary badge	Boolean to badge
validated	ContactInformation	Validated icon	Boolean to checkmark

Calculated Fields

Derived Field	Source Fields	Component	Calculation
fullAddress	All address fields	ContactInformation	Concatenation with formatting
formattedPhone	contactValue (phone)	ContactInformation	(xxx) xxx-xxxx format

Officer Data Mappings

Database Tables: employee , customerOfficerAssignment

API Endpoint: /api/customers/:customerId/officers

Primary Component: Officers.tsx

Direct Field Mappings

Database Column	UI Component	Display Element	Transformation
firstName + lastName	Officers	Officer name	Concatenation
title	Officers	Title	Direct display
position	Officers	Position	Direct display
officerCode	Officers	Officer code	Direct display
department	Officers	Department	Direct display
email	Officers	Email link	Mailto link
phone	Officers	Phone	Phone link
relationshipType	Officers	Relationship type	Label mapping

Notes Data Mappings

Database Tables: note , noteVersion , noteCategory

API Endpoints: /api/customers/:customerId/notes , /api/accounts/:accountId/notes

Primary Components: NotesSection.tsx , NotesTab.tsx , NoteEditorModal.tsx

Direct Field Mappings

Database Column	UI Component	Display Element	Transformation
title	NotesSection	Note title	Direct display
body	NotesSection	Note content	Rich text render
categoryId	NotesSection	Category badge	Joined with noteCategory
importance	NotesSection	Importance indicator	Color mapping
isPinned	NotesSection	Pin icon	Boolean to icon
isDeleted	NotesSection	Deleted state	Conditional styling
createdAt	NotesSection	Created date	Date formatted
createdByEmployeeName	NotesSection	Author name	Direct display
updatedAt	NotesSection	Modified date	Date formatted
versionNumber	NoteVersionHistoryModal	Version number	Direct display

Calculated Fields

Derived Field	Source Fields	Component	Calculation
truncatedBody	body	NotesSection	truncateText() - max length preview
relativeTime	createdAt	NotesSection	getRelativeTime() - "2 days ago"

Calculated/Derived Fields

TotalRelationshipSummary Component

API Endpoint: /api/customers/:customerId/relationship-summary

Component: TotalRelationshipSummary.tsx

Calculated Field	Source	Calculation
totalAssets	All deposit accounts	Server-side sum of positive balances
totalLiabilities	All loan/credit accounts	Server-side sum of negative balances
netWorth	totalAssets , totalLiabilities	assets - liabilities
quarterOverQuarterChange	Historical balances	Current total - 3 months ago total
quarterOverQuarterPercent	QoQ change	(change / previous) * 100

ClientEngagement Component

API Endpoint: /api/customers/:customerId/engagement

Component: ClientEngagement.tsx

Calculated Field	Source	Calculation
transactionCount30Days	Transactions	Count where date > 30 days ago
lastLoginDate	Online banking events	Max login date
loginFrequency	Login events	Count per time period
channelUsage	Login events	Group by channel, count
engagementScore	Multiple metrics	Weighted scoring algorithm

Deposits Component

API Endpoint: /api/customers/:customerId/deposit-analytics

Component: Deposits.tsx

Calculated Field	Source	Calculation
totalBalance	All deposit accounts	Sum of balances
balanceByType	Accounts grouped by type	{ checking: X, savings: Y, cd: Z }
pieChartData	balanceByType	Percentage calculation per type
monthOverMonthGrowth	Trend data	calculateGrowth() function
trendData	Historical balances	Time series for charts

Pie Data Calculation (Deposits.tsx lines 111-134):

```
const getPieData = () => {
  const { checking, savings, cd } = analytics.balanceByType;
  const total = checking + savings + cd;

  return [
    { name: 'Checking', value: checking,
      percentage: total > 0 ? ((checking / total) * 100).toFixed(0) : '0' },
    { name: 'Savings', value: savings,
      percentage: total > 0 ? ((savings / total) * 100).toFixed(0) : '0' },
    { name: 'CD', value: cd,
      percentage: total > 0 ? ((cd / total) * 100).toFixed(0) : '0' }
  ].filter(item => item.value > 0);
};
```

Formatting Functions Reference

Location: client/src/lib/dateFormatters.ts

Date Formatters

Function	Purpose	Example Output
formatDate()	Standard date	"Dec 10, 2024"
formatDateLong()	Long date	"December 10, 2024"
formatTime()	Time only	"2:30 PM"
formatDateTime()	Date + time	"Dec 10, 2024 2:30 PM"
formatTransactionDate()	Transaction date	"12/10/2024"
formatStatementDate()	Statement date	"December 2024"
formatMonthYear()	Month/year	"Dec 2024"
getRelativeTime()	Relative time	"2 days ago"
formatCompactDate()	Compact format	"Dec 10, 2:30 PM"

Currency/Number Formatters

Function	Purpose	Example Output
<code>formatCurrency()</code>	Currency display	"\$1,234.56"
<code>formatPercentage()</code>	Percentage	"4.25%"

Utility Functions

Function	Location	Purpose
<code>calculateAge()</code>	CustomerDashboard, CustomerDetailModal	DOB to age
<code>maskAccountNumber()</code>	AccountCard	"****1234"
<code>getProductName()</code>	AccountCard	Account type to display name
<code>truncateText()</code>	NotesSection	Text preview with ellipsis
<code>safeString()</code>	NotesSection	Null-safe string access
<code>safeNumber()</code>	NotesSection	Null-safe number access

API Endpoints Summary

Endpoint	Component(s)	Data Type
/api/customers/search	CustomerSearch	Search results
/api/customers/:id/details	CustomerOverview, CustomerDetailModal	Customer + related data
/api/customers/:id/accounts	AccountSummaryTableVersion, AccountList	Account list
/api/customers/:id/transactions	TransactionHistory	Transaction list
/api/customers/:id/deposit-analytics	Deposits	Analytics object
/api/customers/:id/relationship-summary	TotalRelationshipSummary	Summary metrics
/api/customers/:id/engagement	ClientEngagement	Engagement metrics
/api/customers/:id/officers	Officers	Officer list
/api/customers/:id/notes	NotesSection	Notes list
/api/accounts/:id/transactions	TransactionHistory	Transaction list
/api/accounts/:id/debit-cards	AccountSummaryTableVersion	Debit card list
/api/households/:id	HouseholdPage	Household details
/api/households/:id/members	HouseholdRelationships	Member list
/api/households/:id/accounts	HouseholdPage	Account list

Document Version

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